

BACKGROUND

Title/Citation Comparing two doses of intramuscular ketorolac for treatment of acute musculoskeletal pain in a military emergency department

<https://www.sciencedirect.com/science/article/abs/pii/S073567572100629X?via%3Dihub>

Background

- Ketorolac tromethamine is a non-steroidal anti-inflammatory drug; its clinical efficacy has been proven for various pain management for Renal Colic, Post Operative, Cancer and Musculo skeletal pain.
- Ketorolac is the most commonly used parenteral analgesics in ED for the treatment of moderate to severe Pain.

GENERAL STUDY OVERVIEW

Purpose

- The purpose of this study was to assess a low-dose versus a high-dose of intramuscular (IM) and Intravenous (IV) ceiling effect of ketorolac for non-inferiority in adults with acute MSK pain in an emergency department (ED).
- There have been studies done for the 10 mg dose of IV Ketorolac to evaluate the ceiling effect for analgesic purpose, but there is less evidence for IM dose of 30 to 60 mg, which is fairly high compared to IV dose.
- In this study the adverse effect of high doses was also evaluated.
- The study was also done to prove cost effective with the prefilled syringes and vials are available for 15 mg and 60 mg dose.

Funding United States Army with no conflict of interest

Design

- This was a single-blind, randomized controlled non-inferiority trial following the guidelines in 2010.
- The sample size adults aged 18 to 55 years with previously diagnosed of Acute MSK pain.
- Patients were blinded to the IM dose administered, but the prescriber was aware of the route and dose to be administered to the patient.

METHODS

Inclusion criteria

- The selected age sample was an 18–55-year-old DoD or VA beneficiary with acute MSK pain reported via Visual Analog Scale (VAS) 20 or greater in the category 4 and 5 in the severity index with duration of less than 30 days.
- Patients were randomly selected only when the attending physician determined the type, dose and route of Ketorolac to be administer.

Exclusion criteria

- Patient weight with less than 50 kg, pregnant, breast feeding, NSAID allergy, history of renal and GI bleeding risk was excluded from the study.
- Patients who required rescue analgesia were also excluded from the studies.
- Patients with history of BP <90 or >180 mmHg with PR <50 or >150/min
- Patient taking OTC or prescribed Non-opioid or Opioid analgesic were also excluded from studies.
- Any anticoagulant and contraindicated with NSAIDs prescription on board was also excluded from the studies automatically.
- A total of 65 patients were excluded from the studies.

Intervention	→ Patients were randomized to choose 15 mg or 60 mg IM dose to admin and were blinded to the dose they received.
Primary outcome	→ The primary outcome was the change in VAS pain score after 60 minutes admin the IM dose of Ketorolac.
Secondary outcomes	→ The secondary outcome was the change in VAS pain score after 30 minutes admin the IM dose of Ketorolac. → The secondary outcome assessing was for dose dependence adverse effects during the time frame of 60 minutes.
Procedures	→ The data was reporting continuous variables as means with standard deviations and ordinal variables as proportions with 95% confidence intervals with statistical significance were defined as $p < .05$. → A minimal clinically relevant difference and non-inferiority margin of 13 mm on the VAS was chosen based on prior studies where subjects annotated at least “a little better” or “a little worse” for change in pain scores. → The standard deviation of minimum sample size of 110 (55 subjects in each arm) when the power is set at 0.8 with an alpha of 0.05. To account for a potential dropout of 10%, 122 subjects were recruited.
Statistical analyses	→ At 60 minutes post admin of 15 mg dose patient reported 29.7 mm of pain reduction vs. 60 mg dose the pain reduction was reported as 29.9 mm, which shows the mean difference of pain reduction of 0.1 mm with 95% CI parameter with p value of 0.98 which was less than predetermine non inferior margin of 13 mm. → At 30 minutes post admin of 15 mg dose patient reported pain reduction mean was 1.7 mm with 95% CI value with $p = 0.63$

→ RESULTS

Baseline characteristics	→ There were total sample of 110 patients with same ratio of male and female with the same range of age. The 15 mg and 60 mg dose were given to 55 patients from each group. → The location of the pain was also considered with the duration of pain for <24 hrs to >24 hours. → The reported adverse effect was reported more in 60 mg dose group with p value of 0.16 (1/9 vs 9/9)
Summary of outcomes	→ At 60 min post administration, those receiving the 15mg dose had a decrease in pain from baseline of 29.7mm and those receiving the 60 mg dose had a decrease in pain from baseline of 29.9 mm. The study's primary outcome of the mean difference between groups at 60-min post administration was 0.1 mm and was less than the predetermined noninferiority margin of 13 mm (Shows in Fig 1 on last page). → Similar results were seen between groups at 30 min post administration with a mean difference of -1.7 mm. (Fig 2 on Last Page) → There were no major adverse effects reported at 60 min and minor adverse effects were more frequent in the 60 mg group with burning at the injection site being the most commonly reported.

AUTHORS' DISCUSSION & CONCLUSIONS

Author's Discussion	→ The study was most closely designed as previous study by Motov et al. where they evaluated IV ketorolac three different (10 mg, 15mg, 30 mg) doses in the treatment of undifferentiated acute moderate to severe pain. → As of our study we administered IM ketorolac and evaluated it for treatment for MSK pain by comparing two doses of ketorolac (15 mg and 60 mg). Two previous studies also evaluated
----------------------------	---

different doses of IM ketorolac for the treatment of cancer and found no statistically significant difference between the compared doses, supporting the efficacy of a lower dose of IM ketorolac. Similar to these studies, we also found a lower dose of IM ketorolac was non-inferior for pain management.

- The pain level was chosen to be 20 mm on the VAS and participation in the study was completely voluntary with the knowledge they would receive an IM injection of a NSAID to help decrease their pain. Furthermore, only 17 subjects (15% of total) reported a VAS score of less than 50 mm.
- The adverse effects reported in this study are dependence not related to the administration of ketorolac. A significantly higher incidence of subjective adverse effects was detected in the 60 mg group than the 15 mg group. Garcia-Rodriguez et al. also demonstrated a significant increase in relative risk for gastrointestinal bleeding with higher doses of IM and oral ketorolac, as well the study of Strom et al. demonstrated clear dose-dependence adverse effects with higher doses of ketorolac.

Author's conclusions

- A 15mg IM dose of ketorolac was found to be non-inferior to a 60mg dose for short-term pain relief of acute MSK pain.
- Subjective adverse effects, while not numerous, were more often reported with the 60 mg dose.
- Discontinuing the practice of ordering 60 mg IM doses of ketorolac in place of a lower dose for acute MSK pain should be considered.

→ PRESENTER'S DISCUSSION & CONCLUSIONS

Study strengths

- Data analysis and sample size calculation was performed using the statistical software SPSS (IBM SPSS Statistics for Windows, Version 25.0. Armonk, NY: IBM Corp.).
- The primary outcome was tested using a one-sided, two-sample equal-variance t-test.
- To assess the lower dose for a difference in adverse effects, a one-sided z-test for two independent proportions were used.

Study limitations

- This study was conducted at a single ED in which all of the study participants were Department of Defense (DoD) or Veterans Affairs beneficiaries and most of them were heterogeneous sampling of the U.S. population.
- Additionally, the exclusion criteria also eliminate the group of subjects who were triaged as ESI 1–3 who believed will be more likely to require stronger therapies for pain control or may have had multi-organ dysfunction and limiting the use of NSAIDs. Another limitation was that the adverse effects in this study were only examined for up to 60 min after the administration of ketorolac, as the well-known adverse effects of ketorolac include renal injury and gastrointestinal bleedings can happen later after the treatment, which were not evaluated in this study.
- This study was designed as a non-inferiority trial where it was not establishing the superiority of the various doses of ketorolac in management of pain and were only able to observe whether a 15 mg dose of IM ketorolac is non-inferior to a 60 mg dose.

Conclusion and Application to patient care

- This study provides strong evidence that a 15 mg dose of IM ketorolac is non-inferior to a 60 mg dose of IM ketorolac for the treatment of acute MSK in adult patients presenting to the ED.
- Discontinuing the practice of ordering 60 mg IM doses of ketorolac in place of a lower dose for acute MSK pain should be considered.

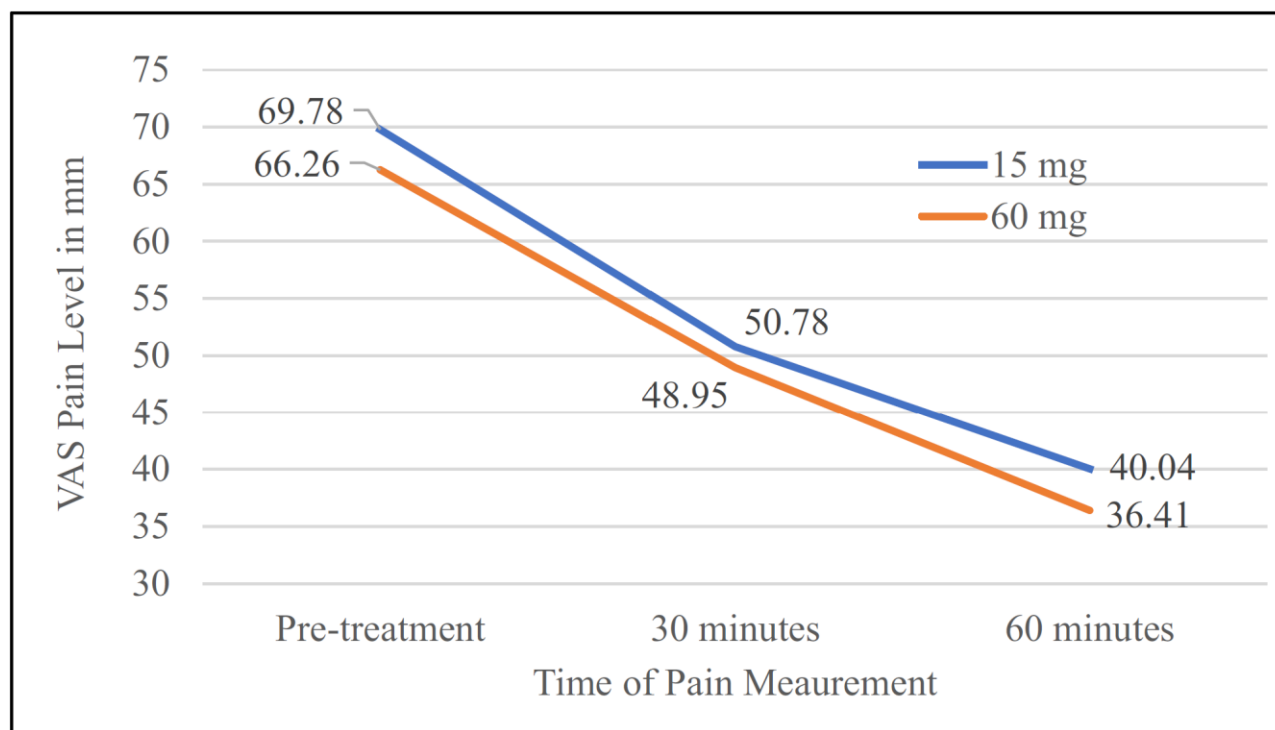


Fig. 1. Reduction in VAS pain (in millimeters) level after single IM injection of ketorolac at 15 mg and 60 mg.

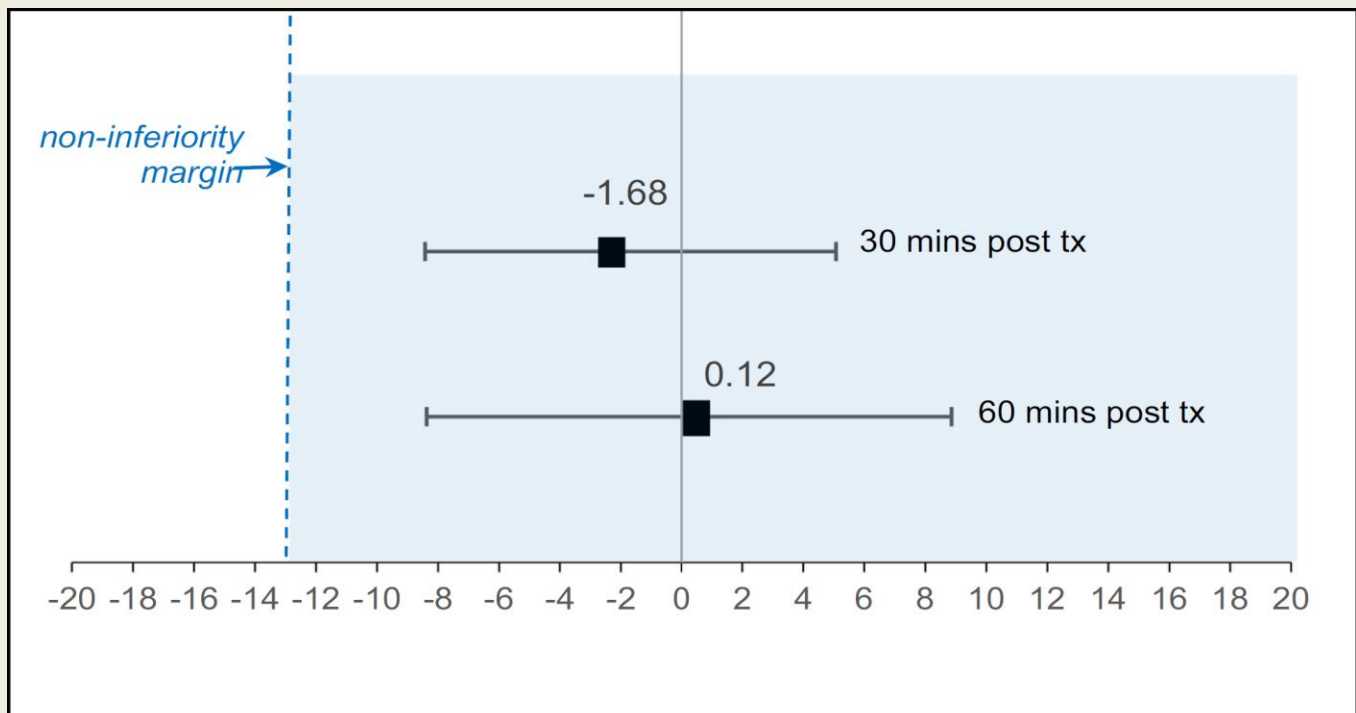


Fig. 2. Mean difference with 95% CI in VAS pain reduction between 15 mg and 60 mg groups at 30 and 60-min. The mean difference is represented by the solid black square with the horizontal bar being the 95% CI.

